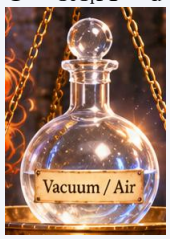
Capacitors — Variables Reference

Capacitance, charging/discharging and AC behaviour: names, symbols, units and the formulas each quantity appears in.

Name	Symbol	Unit	Unit symbol	Common formula(s)	
Capacitance	**C**	farad	**F**	$C = Q \div V$ $\tau = RC$ $C = \varepsilon_0\varepsilon_r A \div d$ $Xc = 1 \div (2\pi fC)$ Series: $1 \div CT = 1/C1 + 1/C2 + \ldots$ Parallel: $CT = C1 + C2 + C3 + \ldots$	
Charge	**Q**	coulomb	**C**	$Q = CV$ $E = \frac{1}{2}QV$	

Name	Symbol	Unit	Unit symbol	Common formula(s)	
Voltage	**V**	volt	**V**	$Q = CV$ $E = \frac{1}{2}QV$	
Energy stored	**E**	joule	**J**	$E = \frac{1}{2}QV$ $E = \frac{1}{2}CV^2$	
Time constant	**τ (tau)**	second	**s**	$\tau = RC$	

Name	Symbol	Unit	Unit symbol	Common formula(s)	
Capacitive reactance	**Xc**	ohm	**Ω**	$Xc = 1 \div (2\pi fC)$	
Impedance	**Z**	ohm	**Ω**	$Z = \sqrt{(R^2 + Xc^2)}$	
Plate area	**A**	square metre	**m²**	$C = \varepsilon_0\varepsilon_r A \div d$	
Plate separation	**d**	metre	**m**	$C = \varepsilon_0\varepsilon_r A \div d$	
Frequency	**f**	hertz	**Hz**	$Xc = 1 \div (2\pi fC)$	

Name	Symbol	Unit	Unit symbol	Common formula(s)	
Permittivity of free space	**ε_0 (epsilon nought)**	farad per metre	**F m^{-1}**	Constant: $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1} $C = \varepsilon_0\varepsilon_r A \div d$ 	
Dielectric constant / relative permittivity	**ε_r (epsilon sub r)**	(no unit — a ratio)	—	$C = \varepsilon_0\varepsilon_r A \div d$	


Exchange Rate
1 F = 10⁻⁶ μF
F
μF